

Early Warning Indicators of NASDAQ Asset Bubbles via Hybrid PSY-LSTM Models

Fatima Sultakeeva, Master student, fatimasultakeeva@jnu.ac.kr

Tae Jong Choi, Professor

Corresponding author. email: ctj17@jnu.ac.kr

**Graduate School of Data Science
Chonnam National University, South Korea**

Content

- 1.** Background
- 2.** Problem statement
- 3.** Goal & Necessity of the study
- 4.** Literature review
- 5.** Study approaches
- 6.** Results
- 7.** Conclusion of research



Background



Problem



Goal



Approaches



Findings



Conclusion

A FINANCIAL BUBBLE

is a situation where the price of an asset rises too quickly and far beyond its fundamental value.

This price surge is often driven by speculation, hype, and fear of missing out (FOMO). Eventually, the market "wakes up", demand disappears — and prices crash. This collapse is known as the bursting of the bubble.

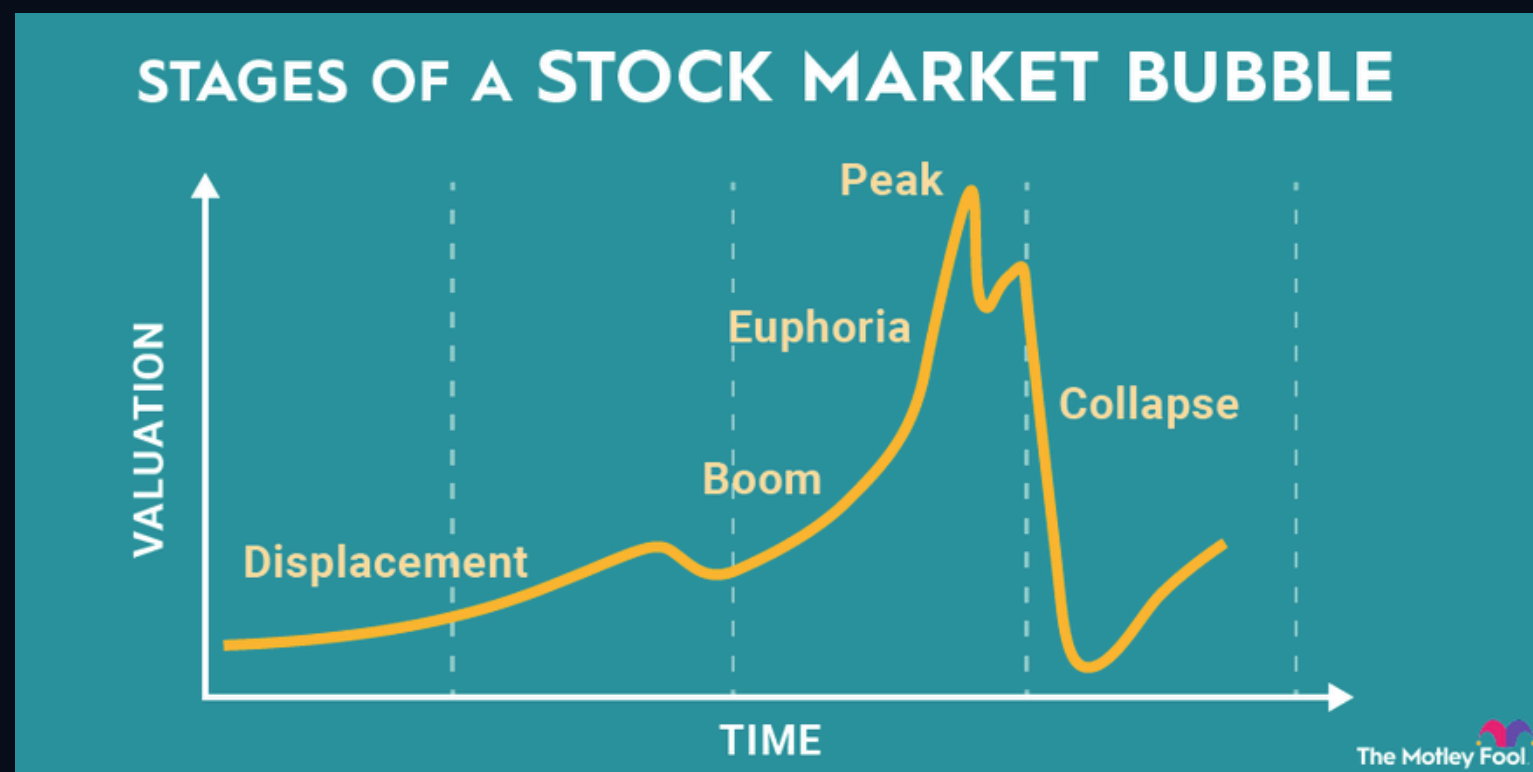


Image source: The Motley Fool

1. **Displacement:** A new factor or policy creates investment opportunity.
2. **Boom** Prices rise rapidly, attracting new investors.
3. **Euphoria** Investors make irrational decisions ("Greater Fool Theory")
4. **Peak** Prices top out, "smart money" begins to sell.
5. **Collapse** Prices fall sharply, financially ruining retail investors.

Major Historical Asset Bubbles and Their Consequences

<i>Name</i>	<i>Period</i>	<i>Asset</i>	<i>Consequences</i>
<i>Tulip Mania</i>	<i>1636–1637</i>	<i>Tulip bulbs</i>	<i>Prices rose ~100× then collapsed ~99%; widespread bankruptcies among speculators.</i>
<i>South Sea Bubble</i>	<i>1719–1720</i>	<i>British joint-stock companies</i>	<i>Extreme overvaluation followed by collapse; widespread investor losses; early systemic shock.</i>
<i>Dot-com bubble</i>	<i>1999–2001</i>	<i>Internet companies</i>	<i>~\$5 trillion lost from NASDAQ; massive tech-sector crash; thousands of companies failed.</i>
<i>U.S. housing bubble</i>	<i>2006–2008</i>	<i>Real estate, mortgages</i>	<i>Triggered the Global Financial Crisis; bank failures; deep recession; worldwide contagion.</i>
<i>Cryptocurrency bubble</i>	<i>2017</i>	<i>Bitcoin & altcoins</i>	<i>Over 80% price collapse; extreme volatility; retail-investor-driven boom and bust.</i>



These historical episodes show that bubbles recur across centuries and markets, highlighting the need for early-warning systems.



Background



Problem



Goal



Approaches



Findings

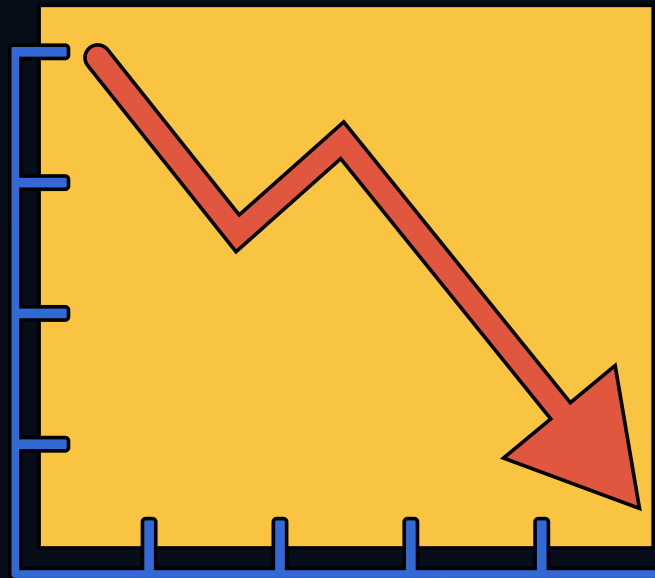


Conclusion

THE FINANCIAL BUBBLE

create systematic financial risk that can affect people's lives indirectly through various mechanisms.

Their consequences include:



-  Stock price collapse , losing money
-  Liquidity crisis
-  Market-wide downturn
-  Loss of investor's trust
-  Economic recession

Existing models react only after the event has already occurred.

Econometric bubble tests can detect bubble periods, but they do not reveal which market indicators contribute to the bubble formation. Therefore, an **Early Warning Indicator (EWI) system** is needed to capture pre-bubble market dynamics.

EWI

EARLY WARNING INDICATORS

are signals or data points that provide an early notice of a potential problem before it becomes a major crisis. They are used in various fields, such as finance, education, and risk management, to allow for timely intervention and mitigation. Examples include a student's failing grades or a bank's declining liquidity metrics.



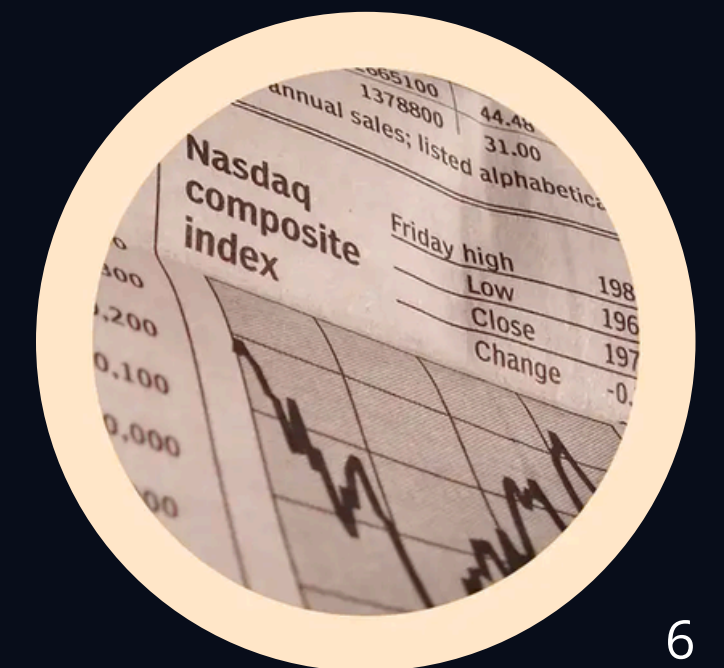
Necessity of EWI

Bubbles have a wide economic impact. Early detection of market indicators can benefit:

- Regulators — for taking preventive actions or mitigate consequences
- Investors — by reducing financial losses
- Researchers — for understanding the market behavior

Why NASDAQ?

- Represents the global technology sector
- Highly volatile and historically bubble-sensitive
- Critical during the new AI boom era





Background



Problem



Goal



Approaches



Findings



Conclusion

The goal is to develop an Early Warning Indicator (EWI) system for the NASDAQ market using a hybrid PSY-LSTM approach, capable of detecting bubble dynamics before they emerge.

Research gaps

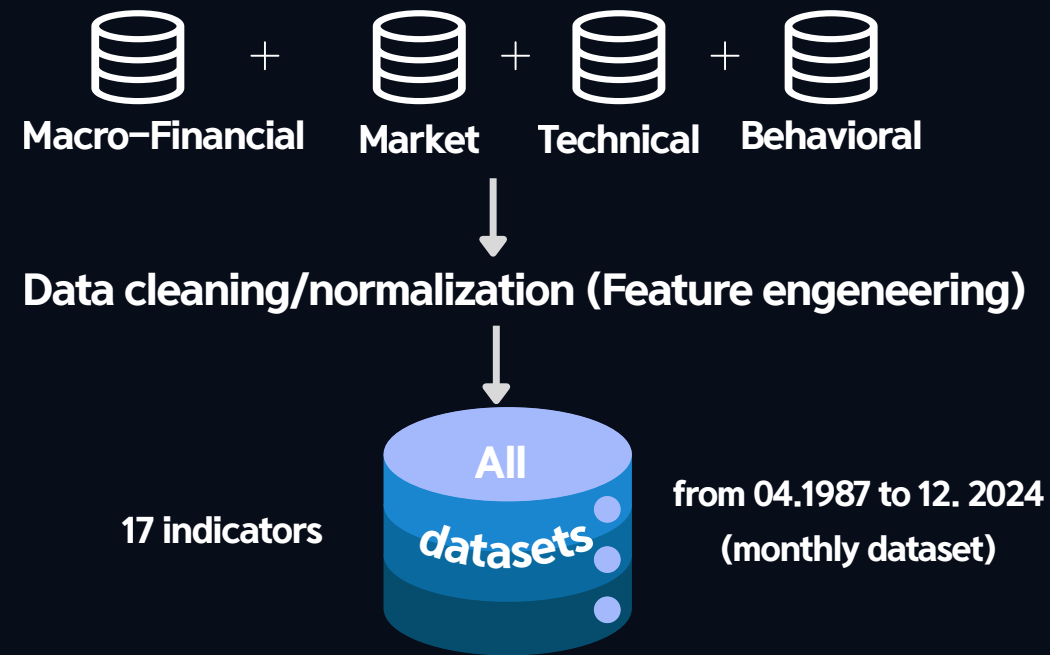
Previous research has mainly focused on banking crises, credit cycles, housing markets, and commodity or metal bubbles, with only limited attention given to stock-market bubbles — especially in technology-driven indices like NASDAQ.

Research Questions

1. How can econometrical test (PSY) date historical NASDAQ bubbles?
2. Does a hybrid model PSY-LSTM outperform standard model within imbalanced data?
3. Which indicators of market act as early-warning signals?

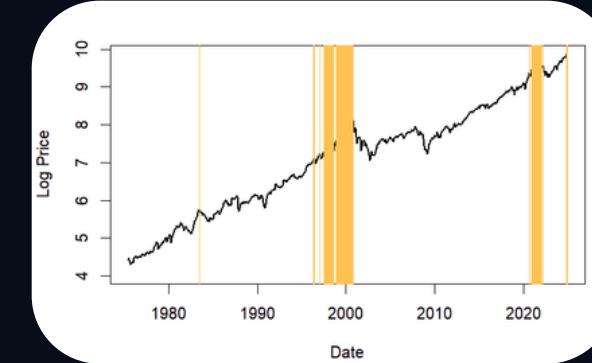


1 Data preparation



2 Target definition

Hybrid model: Stage 1



PSY-Monitoring test (Phillips – Shi & Yu, 2020)

Handbook of Statistics, Volume 42 (2020)

It identifies periods of "explosive" growth in asset prices, which is a key characteristic of a financial bubble.



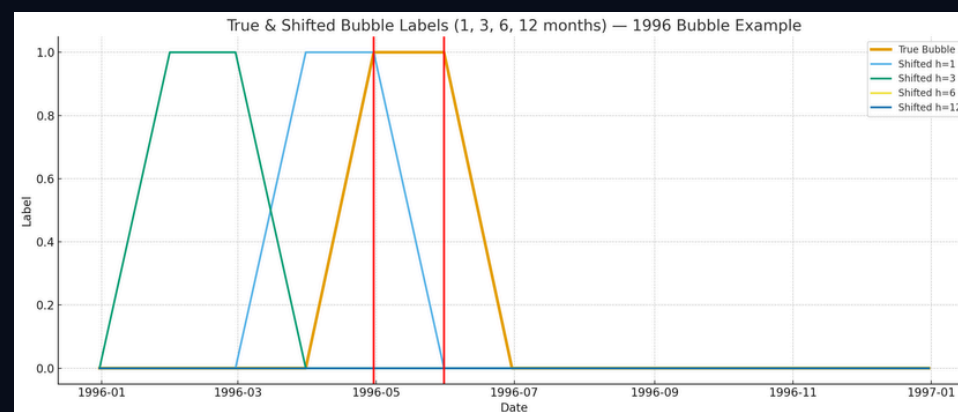
True bubble periods for target labels (Y)

3 EWI system:

Target preparation

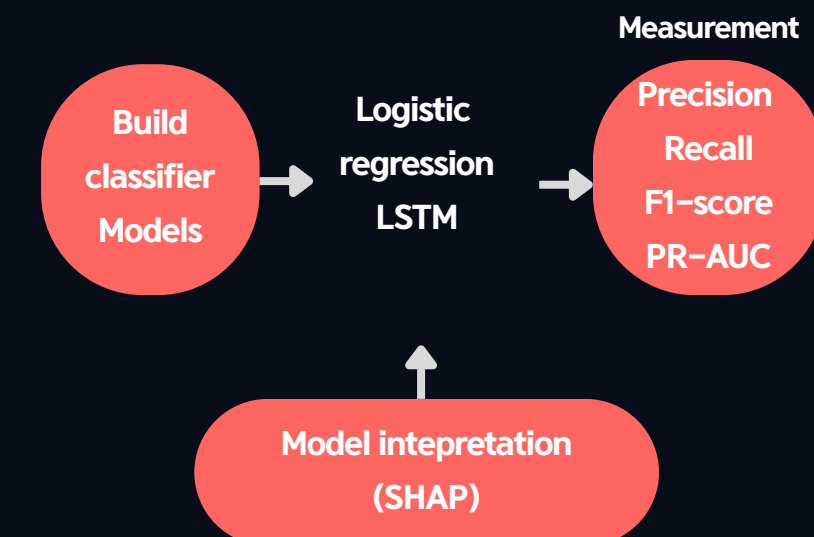
EWI system :

Bubble labels are shifted by 1/3/6/12 months to create early-warning targets before the bubble onset.



4 Hybrid model:

Stage 2





Background



Problem



Goal



Approaches



Findings



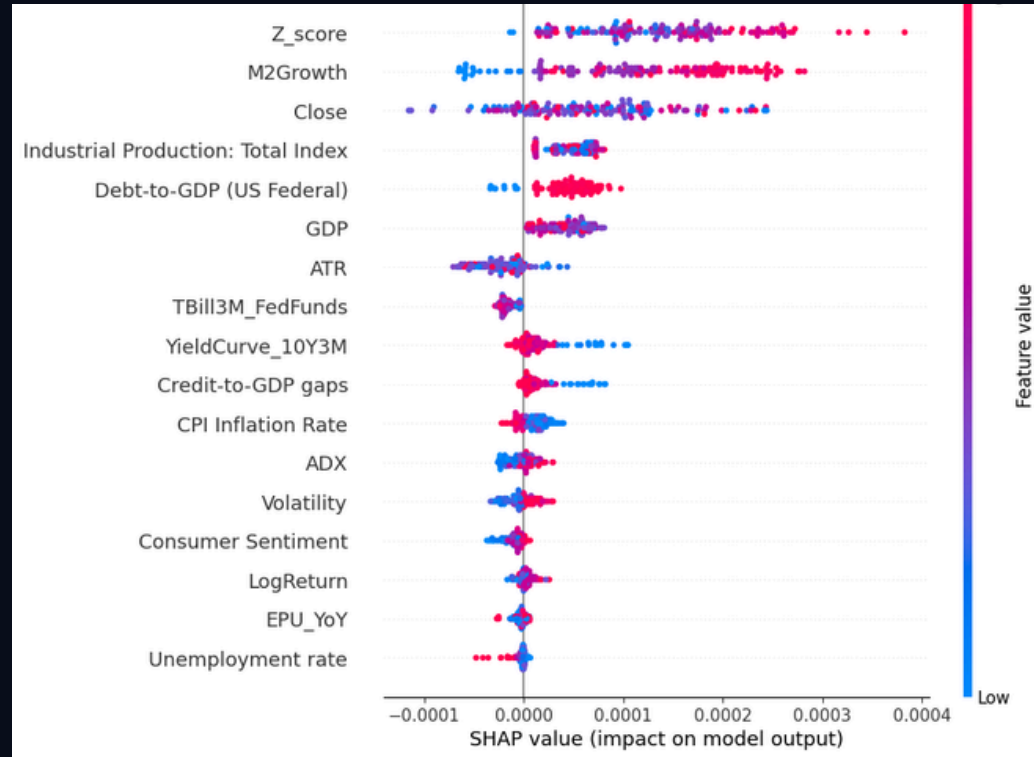
Conclusion

Baseline and hybrid model performance

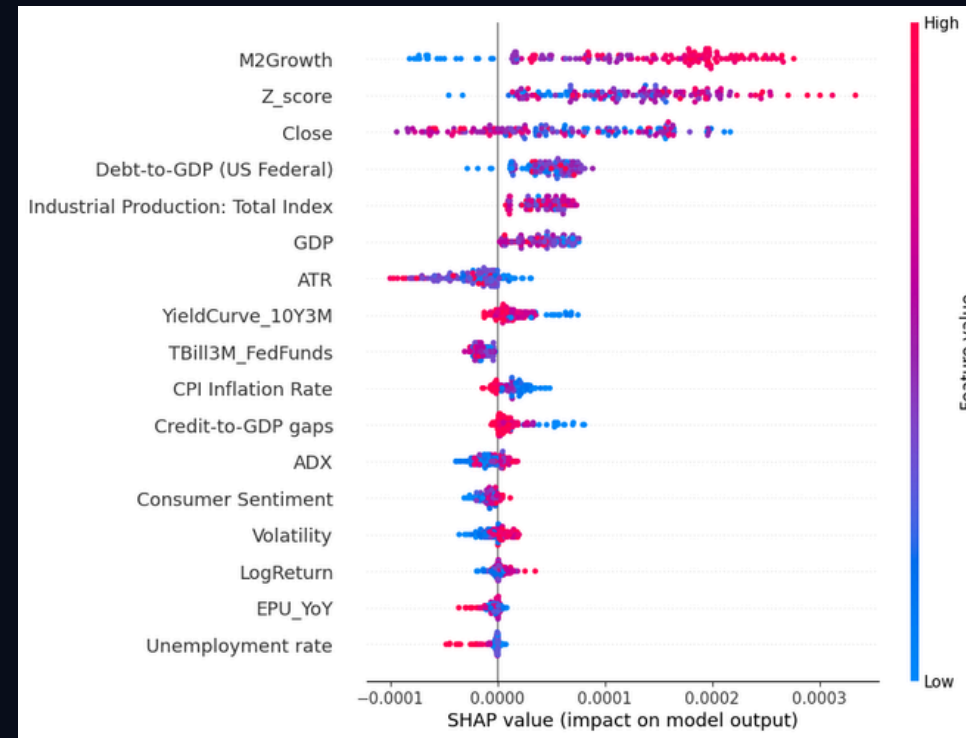
Model	Time horizon	Precision		Recall		F1-score		PR-AUC (Bubble class)
		Normal	Bubble	Normal	Bubble	Normal	Bubble	
LR	1 month	1.00	0.26	0.24	1.00	0.39	0.42	0.65
PSY - LSTM		0.85	1.00	1.00	0.62	0.92	0.77	0.97
LR	3 month	0.89	0.41	0.73	0.68	0.80	0.51	0.68
PSY - LSTM		0.92	1.00	1.00	0.81	0.96	0.90	0.95
LR	6 month	0.83	0.71	0.97	0.26	0.89	0.38	0.56
PSY - LSTM		0.90	0.92	0.97	0.75	0.94	0.83	0.91
LR	12 month	0.83	0.31	0.74	0.42	0.78	0.36	0.47
PSY - LSTM		0.95	0.70	0.86	0.88	0.90	0.78	0.57

SHAP graphs (Hybrid model)

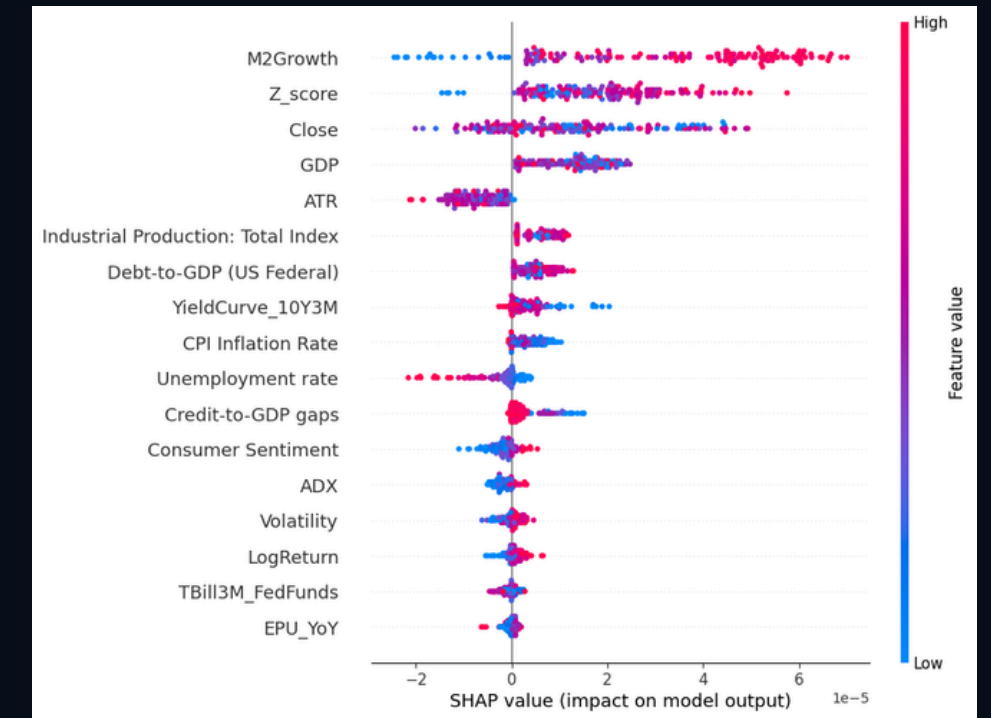
1-month forecasting horizon



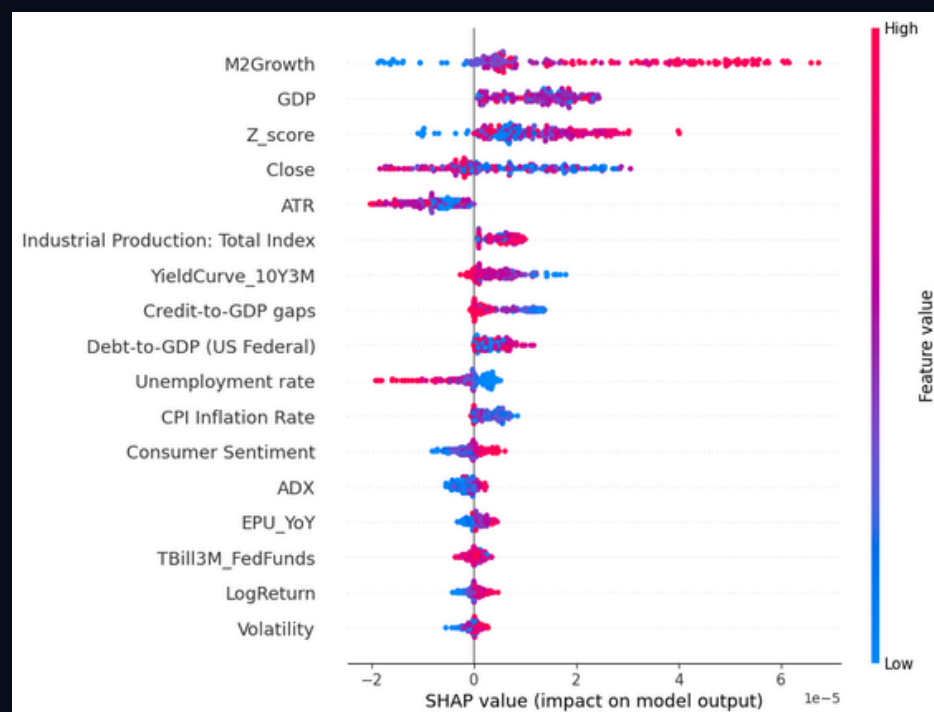
3-month forecasting horizon



6-month forecasting horizon



12-month forecasting horizon



SHAP analysis shows that the most stable early indicators of NASDAQ bubble formation are **M2 Growth, Z-score, Close (price level), GDP, and ATR**. These variables consistently exhibit the strongest influence on the LSTM model's predictions across all horizons (1-12 months).

The most consistent leading indicator across all forecasting horizons in Nasdaq market - **M2Growth**.
High M2 growth → excess liquidity → capital inflow → bubble formation.



Background



Problem



Goal



Approaches



Findings



Conclusion

1. **Hybrid PSY-LSTM provides an effective Early Warning Indicators system for NASDAQ bubble risk.**
2. **It captures nonlinear pre-bubble dynamics that classical econometric and tree based ML models fail to detect.**
3. **6-month forecasting horizon demonstrates the best precision-recall balance for true early warning.**
4. **The model remains interpretable, with SHAP highlighting the role of liquidity, volatility signals.**
5. **This approach supports policy, risk management, and strategic investment decisions.**

KEY TAKEAWAYS

1. Historical NASDAQ bubbles can be detected months before they fully emerge by PSY - LSTM model.
2. SHAP identifies liquidity expansion, volatility, as the key early indicators of bubble formation.
3. The proposed framework is flexible and can support long-term strategic investors, asset managers, policymakers who need forward-looking risk signals



FUTURE WORKS

1. **Genetic Programming (GP)** - train GP models to evolve interpretable early-warning formulas and evaluate their generalization across different markets.
2. **Cross-Market Extension** - apply the pipeline to KOSDAQ, Nikkei 225, S&P500, MOEX, crypto, and emerging markets.
3. **Crash-Based Target** - redefine the target as a crash event (e.g., $\geq 15\text{--}20\%$ drawdown) to build a unified bubble-crash early-warning framework.

THANK YOU!

Fatima Sultakeeva, Master student, fatimasultakeeva@jnu.ac.kr

Tae Jong Choi, Professor

Corresponding author. email: ctj17@jnu.ac.kr

Graduate School of Data Science
Chonnam National University, South Korea

APPENDIX

1 step

PSY-Monitoring (Phillips – Shi & Yu, 2020)

It identifies periods of "**explosive**" (**exponential**) **growth** in asset prices, which is a key characteristic of a financial bubble.

How It Works:

It analyzes financial data using a "rolling window" approach. In each window, it performs a specialized statistical test to check for growth that is too rapid to be normal. When the test statistic consistently exceeds a certain threshold, it signals the presence of a bubble.

Key Idea:

- It's based on recursive right-tailed ADF regressions.
- Instead of testing only one window, the PSY test:
- Rolls through different time windows of the data.
- Computes ADF statistics for each subsample.
- Looks for periods where the statistic exceeds a simulated critical value → indicating explosive root (bubble).

- **Right-tailed ADF (used in PSY):**

Null: the series has a unit root.

Alternative: the series is **explosive** (root > 1, instead of < 1).

👉 So the PSY test is essentially running *many right-tailed ADF regressions* across rolling windows. It checks if the autoregressive coefficient ϕ in

$$y_t = \phi y_{t-1} + \varepsilon_t$$

Real time monitoring of asset markets:

Bubbles and crises.

is **greater than 1** (explosive), rather than just testing if $\phi < 1$.

In: Handbook of Statistics, Vol. 42, pp. 61–80. Elsevier.

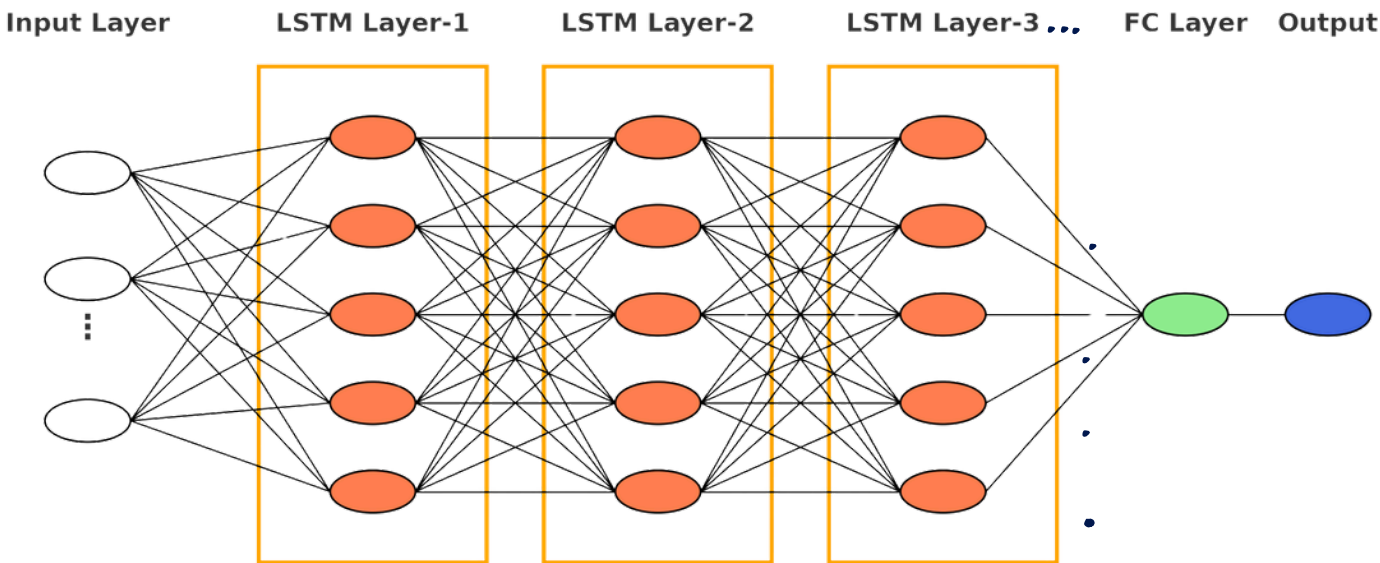
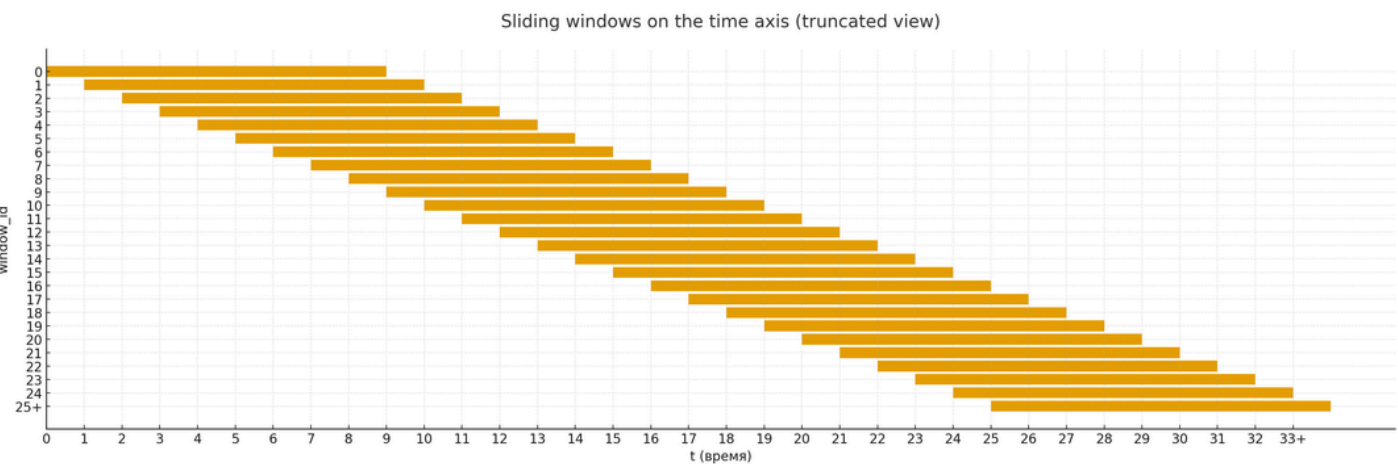
2 step

LSTM Classifier

An LSTM (Long Short-Term Memory) network is a type of recurrent neural network designed for analyzing sequential data, such as text, speech, or financial time series.

Its key feature is the ability to "remember" important information over long periods, which solves the "vanishing gradient" problem. This makes it an ideal tool for forecasting and pattern recognition where the order of data is critical.

Figure 3. Windows for LSTM model



Model uses a 9-month sliding window to capture temporal patterns in market data.(Batch size =32)
By employing shifted labels, we train the model to detect a "bubble" event within the next 3/6/9/12 months to identify critical pre-bubble market signals.

Settings
All selected features were scaled
5 hidden layers: LSTM ×5
Hidden size = 128, dropout = 0.2
Head: Linear 128→1 + Sigmoid
BCEWithLogitsLoss (pos_weight = Neg/Pos)
Adam (lr = 5e-4), batch size = 32
Split 60/20/20, Epochs: 50; Early stopping.

2 step

Data collection

453 months
from
1987 April - 2024
December 60 months -
bubble
393 months - normal

Monthly datasets of Nasdaq: macroeconomic, market, technical (17) for NASDAQ (1987–2024) + bubble label

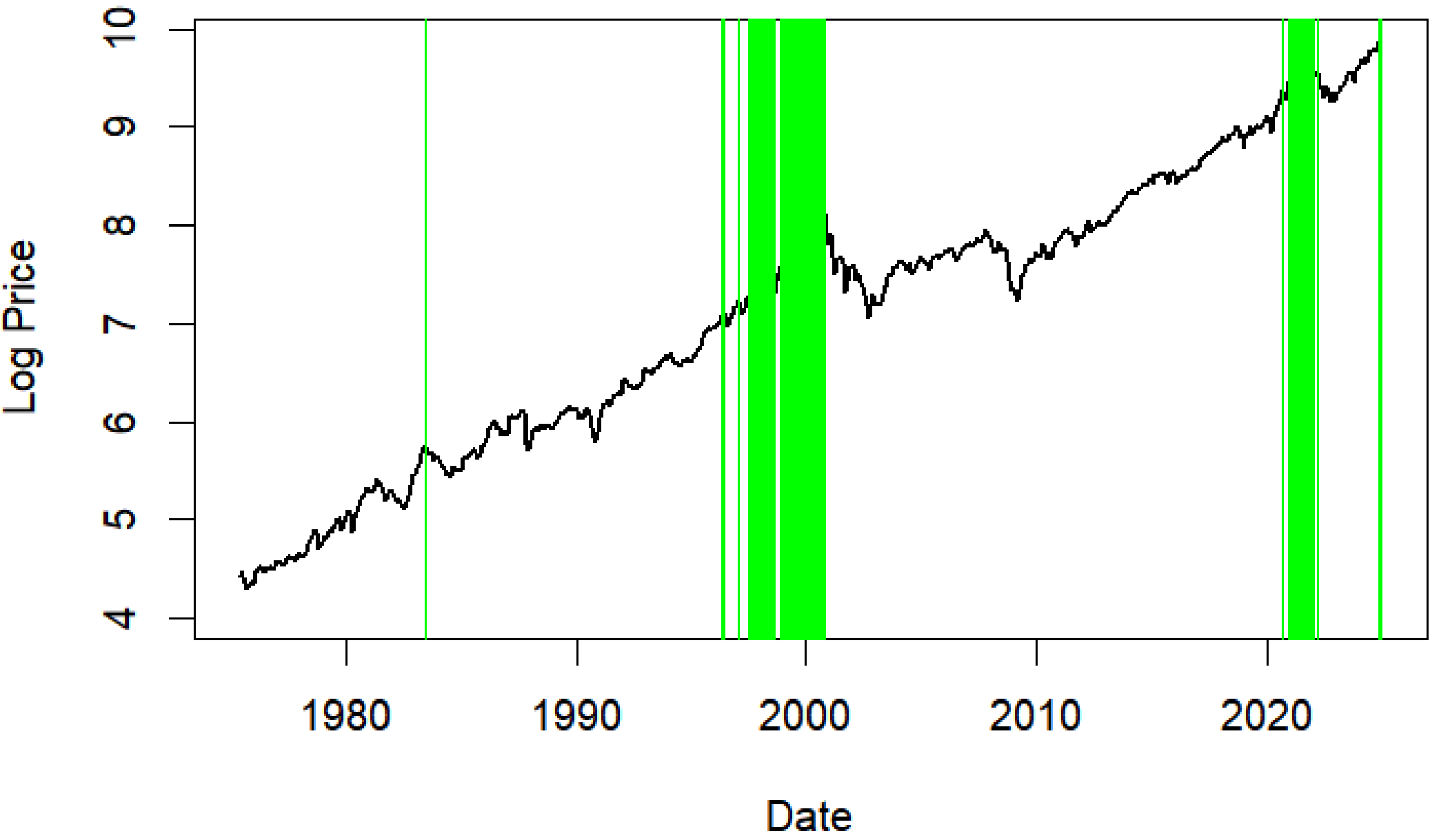
Indicators	Datasets
Macroeconomical indicators	Unemployment rate, GDP, Consumer Sentiment, Credit-to-GDP gaps, M2Growth (money supply expansion), CPI Inflation Rate, Industrial Production: Total Index, Debt-to-GDP (US Federal)
Monetary & financial conditions	YieldCurve_10Y3M, TBill3M_FedFunds
Market-based indicators	Close, LogReturn
Technical indicators	Volatility, Z_score, ADX, ATR
Behavioral / uncertainty indicators	EPU_YoY

Results of PSY tests

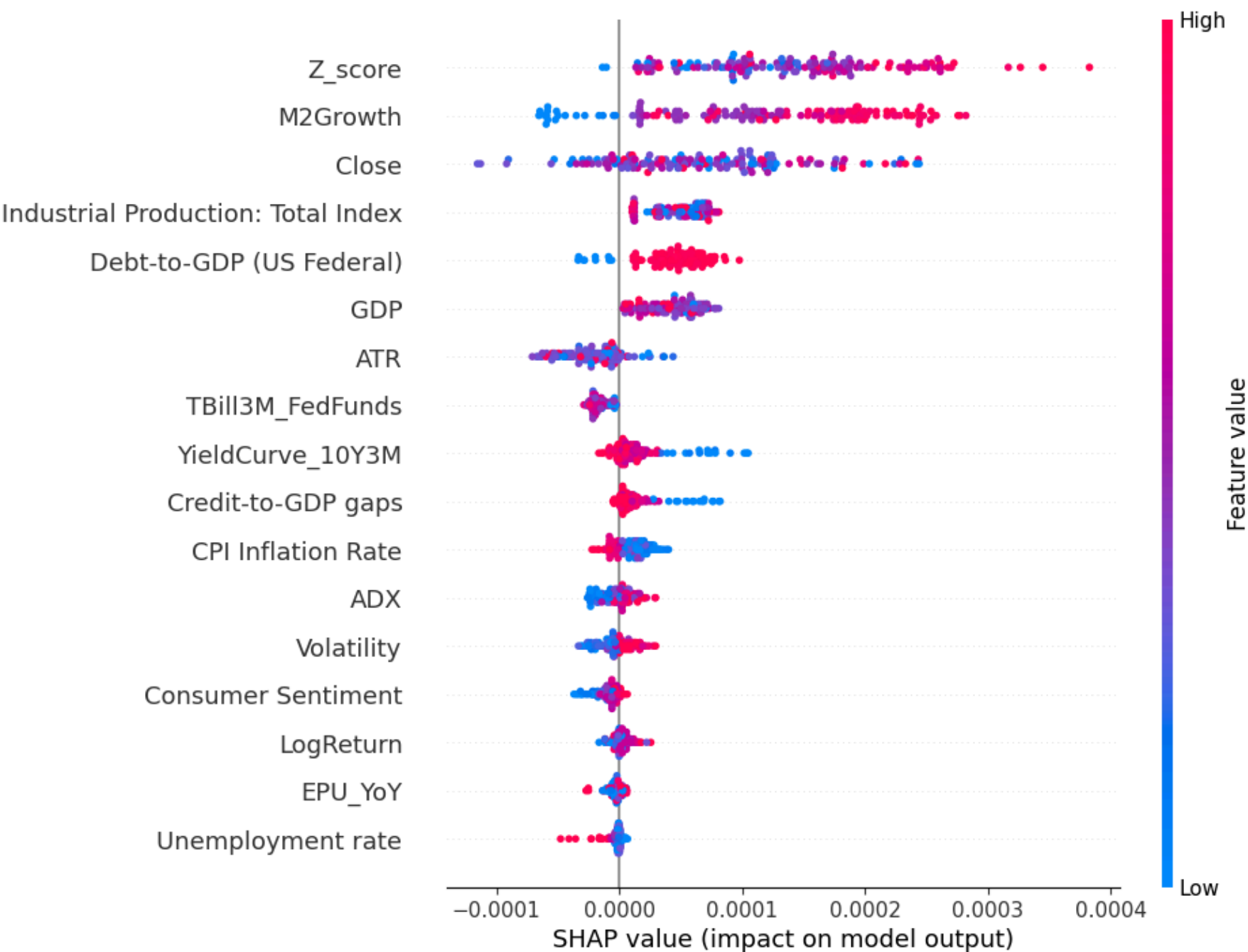
Bubble dates

NASDAQ monthly log price

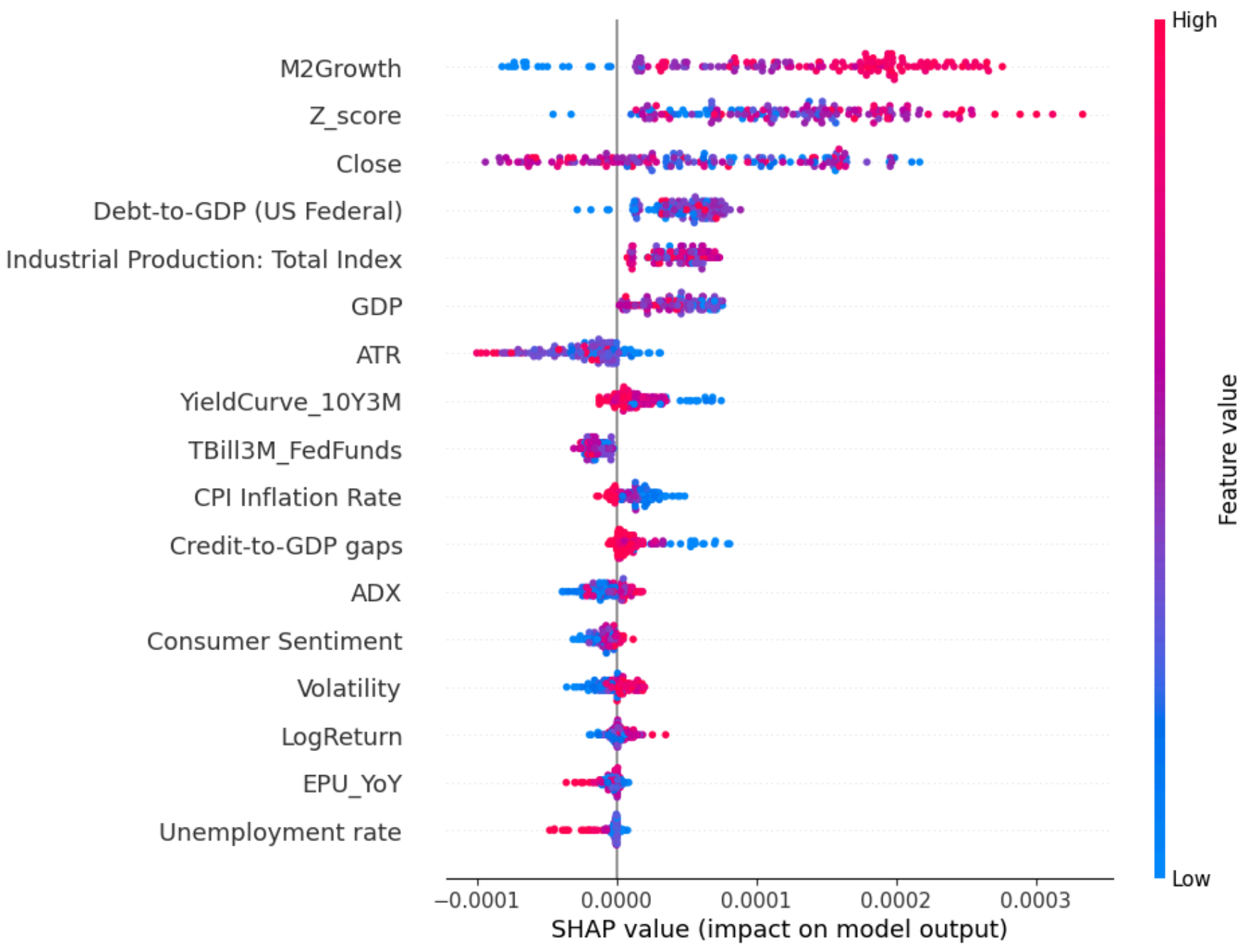
	start	end
1	1983-06-30	1983-06-30
2	1996-04-30	1996-05-31
3	1997-01-31	1997-01-31
4	1997-06-30	1998-07-31
5	1998-11-30	2000-10-31
6	2020-08-31	2020-08-31
7	2020-11-30	2022-01-31
8	2022-03-31	2022-03-31
9	2024-11-30	2024-12-31



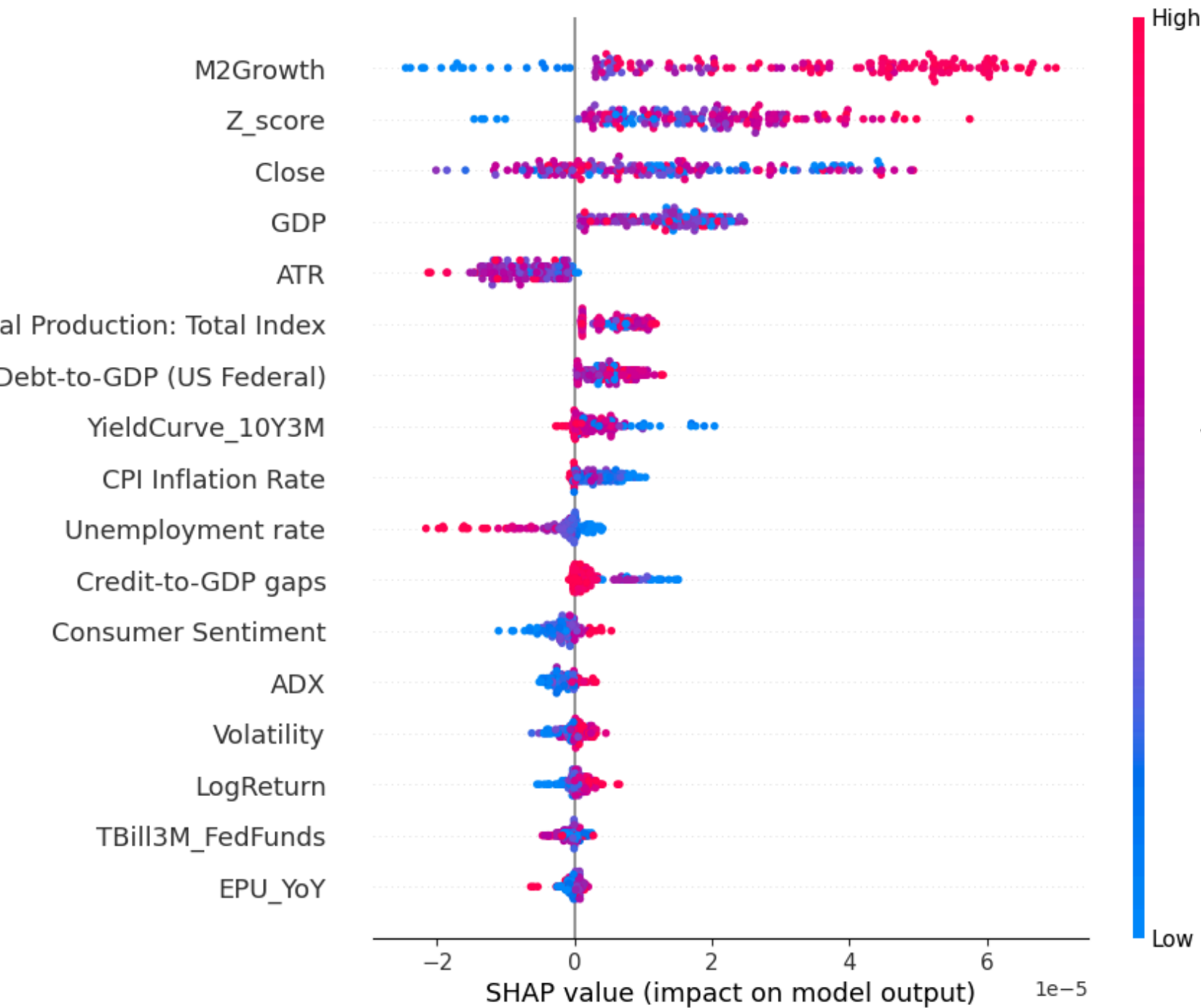
Early indicators before a 1-month forecasting horizon



Early indicators before a 3-month forecasting horizon



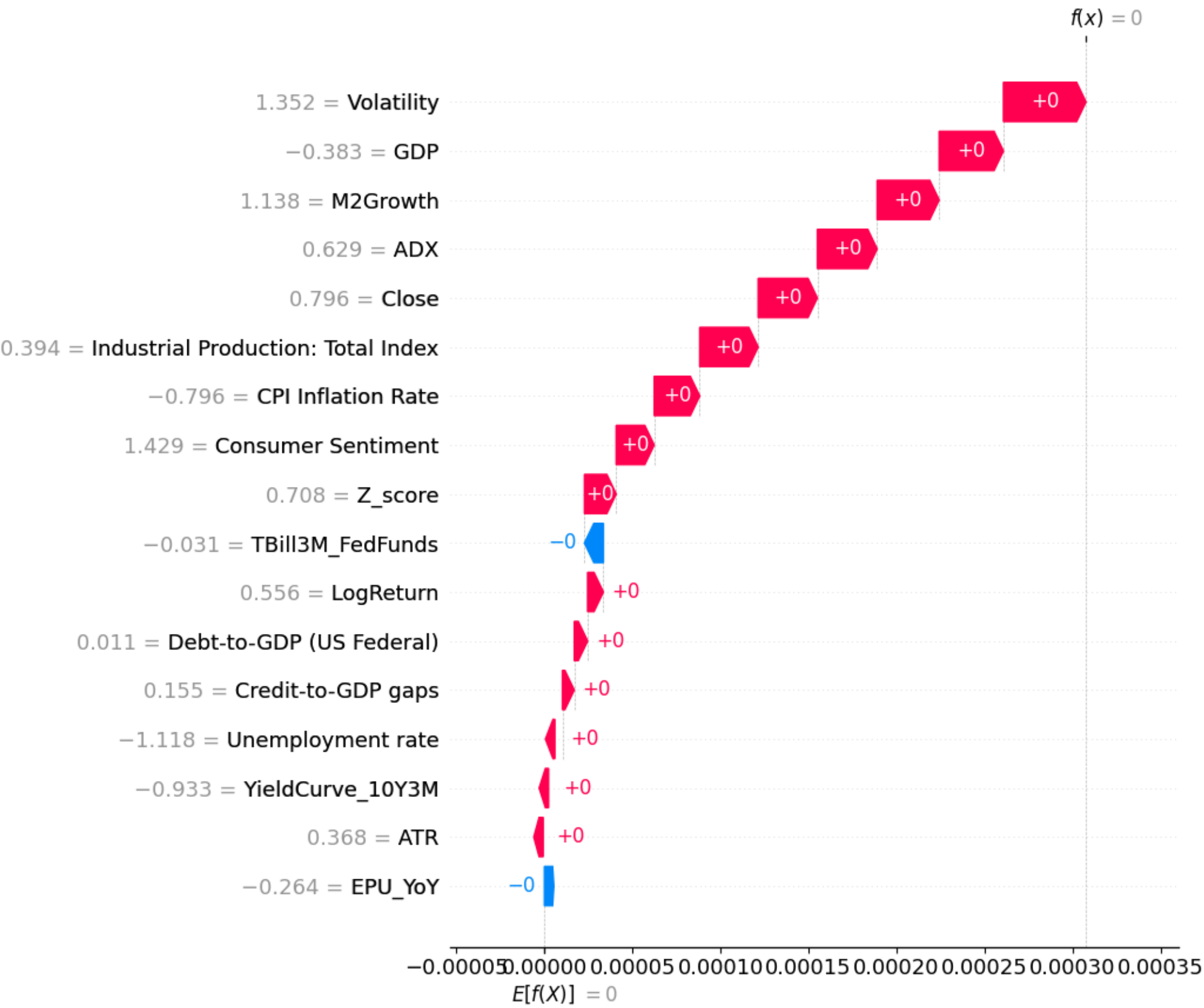
Early indicators before a 6-month forecasting horizon



Early indicators before a 12-month forecasting horizon



Example Dotcom bubble before 3 month of bubble formation



Top Positive Contributors (Bubble Probability ↑)

These factors increase the probability of a bubble:

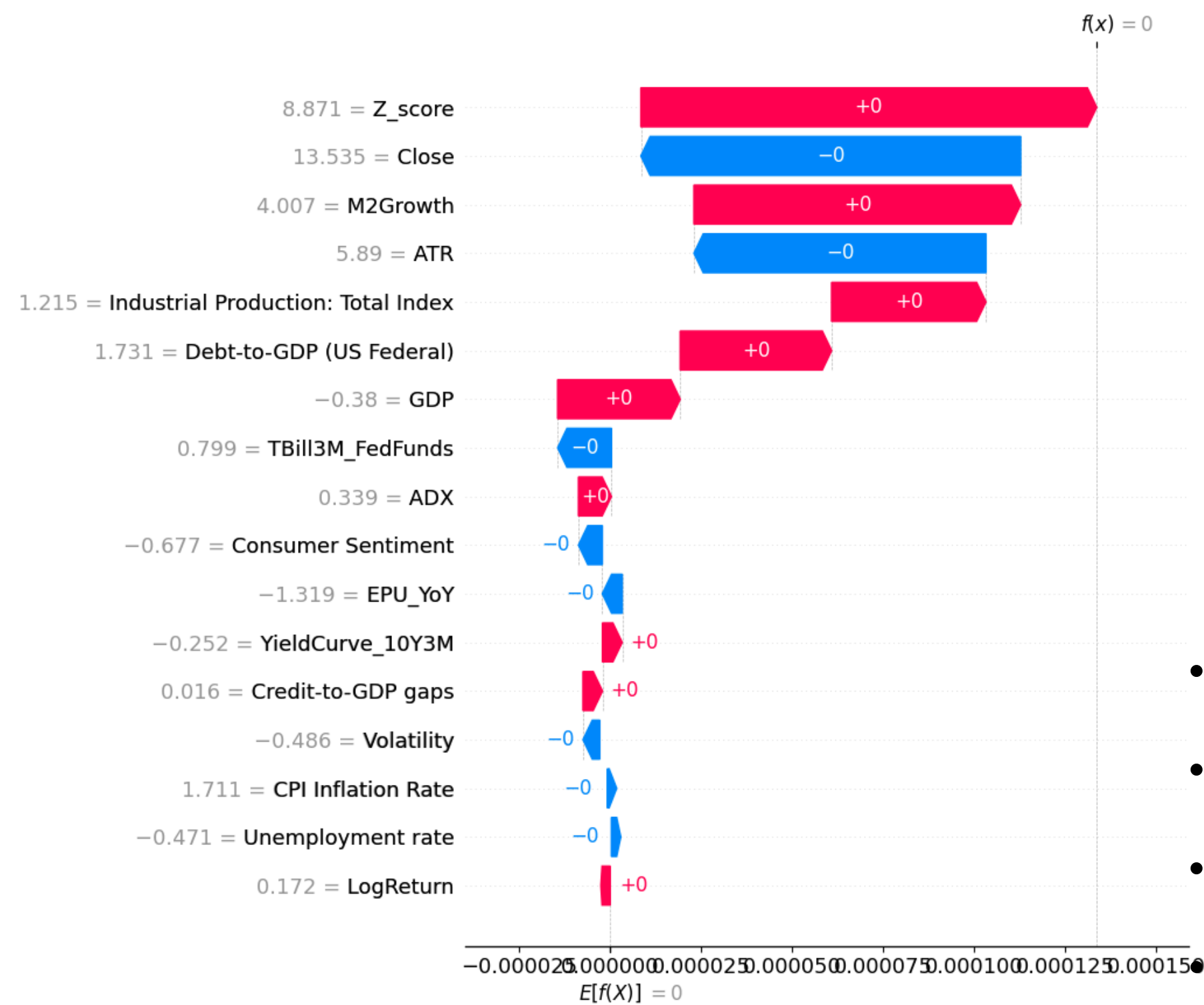
- Volatility — strong market instability
- M2 Growth — excess liquidity and rapid money expansion
- ADX — strong trend momentum before price acceleration
- Close price — fast rising NASDAQ levels
- Consumer Sentiment — optimistic expectations and speculative behavior
- Industrial Production — strong macro expansion
- ATR — increasing price range and market stress

Negative Contributors (Bubble Probability ↓)

These factors reduce the probability of a bubble:

- CPI Inflation Rate — stable price environment
- YieldCurve_10Y3M — healthy monetary conditions
- Unemployment rate — strong labor market
- EPU_YoY — low uncertainty
- TBill3M_FedFunds — no short-term funding stress
- Debt-to-GDP (US Federal) — stable leverage
- Credit-to-GDP gaps — no excessive credit cycle pressure

Example AI recent bubble before 3 month of bubble formation



Top Positive Contributors (Bubble Probability ↑)

These factors increase the probability of a bubble:

- Z-score - strong price overextension relative to long-term trend
- Close (price level) - rapid NASDAQ appreciation during the AI-driven rally
- M2 Growth - excess liquidity following post-COVID monetary expansion..
- ATR (range-based volatility) - wide price swings indicating rising market instability.
- Industrial Production - macro expansion consistent with overheating conditions.
- Debt-to-GDP (US Federal) - increased government financing providing an accommodative background.

Together, these factors created a “market overheating” profile

Negative Contributors (Bubble Probability ↓)

These factors reduce the probability of a bubble:

- GDP (slightly negative) - the economy was growing steadily, providing a stabilizing effect.
- TBill3M – FedFunds (short-term funding pressure) - did not indicate strong stress at that moment.
- Consumer Sentiment - households were less optimistic than financial markets.
- EPU_YoY (policy uncertainty) - uncertainty was declining, reducing perceived risk.
- YieldCurve 10Y–3M - the curve was not yet fully inverted.
- Volatility - was slightly decreasing, contributing to a lower risk signal.